

BLOCK 3
RURAL DEVELOPMENT MANAGEMENT



UNIT 9 MANAGEMENT OF RURAL DEVELOPMENT PROJECTS

Structure

- 9.0 Objectives
- 9.1 Introduction
- 9.2 Management Concept
- 9.3 Process of Management
- 9.4 Implementation of Rural Development Projects
- 9.5 Let Us Sum Up
- 9.6 Key Words
- 9.7 Suggested Readings

9.0 OBJECTIVES

The aim of this unit is to explain the concept and process of management. An attempt has also been made to examine the scope for use of management techniques in rural development projects.

After completing this unit, you should be able to:

- state the concept and scope of rural development management;
- differentiate between management and administration;
- describe the process of management; and
- identify the factors affecting implementation of projects.

9.1 INTRODUCTION

No matter how systematically a project is planned, there would still be deviations between what is intended and what is actually achieved. The deviations may be in terms of time schedule or resource use or flow of benefits from the project. The most common cause identified for such deviations has been poor management.

Naturally, a series of questions come up when a term like management is used. For example, what is management? How does one define it? How do we differentiate between good management and bad management?

The present unit attempts to explore these questions. An examination of the process of management has also been made to identify measures for better management of rural development programmes.

The underlying basic premise of the unit is that the role of management in our society is crucial for the progress of human beings. It helps to fulfill a great need of our time, viz., to improve standards of living for all people through effective utilization of human and material sources.

1.2 MANAGEMENT CONCEPT

Management is a universal process. It is not merely restricted to a factory, shop or office. It is an operative force in all organizations and tries to achieve stated objectives. Management is, therefore, as much necessary for business firms as for government enterprises and departments.

Management is a distinct type of activity primarily responsible to get things done by directing the efforts of other people. Managers in any field of human effort perform typical managerial functions irrespective of what they are managing. It is difficult to achieve the best possible allocation and utilization of human, material and financial resources without efficient management.

There are several definitions of management. We give here some that are widely used definitions. According to Peter Drucker, the famous management expert: "Management is a multi-purpose organ that manages a business, manages manager, and manages worker and work."

According to Harold Koontz, "Management is the art of getting things done through and with people in formally organized groups. It is the art of creating an environment in which people can perform as individuals and yet cooperate towards the attainment of group goals. It is the art of removing blocks to such performance, a way of optimizing efficiency in reaching goals."

Stoner and Wenkel have defined management as the "process of planning, organizing, leading and controlling the efforts of organization members and of using all other organizational resources to achieve related goals."

The definitions do not materially differ except in terms of emphasis. They indicate that management is concerned with attainment of group goals in an efficient way through the combined efforts of people. This involves working with and directing people and also involves planning, organizing, leading and controlling.

EFL Brech says that management is concerned with seeing that the job gets done –its task put emphasis on planning and guiding the operatives that are going on the enterprise. A. Das Gupta defines management is the creation and control of technological and human environment of an organization in which human skill and capacities of individuals and groups find full scope for their effective use in order to accomplish the objective for which an enterprise has been set up. It is involved in the relationships of the individual, group, organization and the environment.

Most management experts consider management as primarily an art. However, many experts also claim that it has the elements of a science as well. As Koontz and O'Donnell have stated: "Managing like other practices... is an art. It is 'know-how'. It is doing things in the light of the realities of a situation. But the practice of managing, like other practice, will be better by making use of underlying organized knowledge; and it is this knowledge, whether crude or advanced, whether exact or inexact, that, to the extent it is well organised, clear and pertinent, comprises a science. Thus, managing as practice is art; the organised knowledge, underlying it may be referred to as a science."

Administration and Management

Is there any difference between management and administration? There is no unanimity whether the two terms are different or connote the same thing. According to one viewpoint, which is widely held, administration refers to the higher and policy-determining levels. It is concerned with the determination of objectives, policies and organisational structure and strategies. The coordination of finance, production and distribution are its main responsibility. It also acts as the central controlling authority in the organisational set-up. According to this view, administration is higher than management in terms of the conceptual function.

We can summarise the differences as follows:

Administration

Ensures effective direction

Devises organisation

Defines goals

Coordinates finance, production and distribution

Provides guidance, leadership and control

Management

Ensures effective execution

Uses the organisation (Organisation is the medium to exercise managerial functions)

Attempts to achieve these goals

Coordinates activities of others to fulfill an objective

Gets things done by working with people, uses other resources in an organised manner

Not all experts, however, agree that administration is above management. Some are of the view that, in practice, administration and management do not differ. They involve use of similar methods to achieve their objectives. Another view is that management is a much wider term and includes administration. According to this view “Management is the generic term for the total process of executive control involving responsibility for effective planning and guidance of the operations of an enterprise.

Administration is carrying out of the procedures by which the programme is laid down and communicated and the progress of activities is regulated and checked against plans.”

Check Your Progress I

Indicate whether the statements below are true or false by marking in the relevant box.

	True	False
a) Management applies only to commercial undertakings.	☐	☐
b) Management ensures effective execution.	☐	☐
c) Management improves performance by helping to achieve objectives by optimizing efficiency.	☐	☐
d) Management is a science and not an art.	☐	☐
e) Management defines goals, while administration attempts to achieve them.	☐	☐

1.3 PROCESS OF MANAGEMENT

The foregoing discussion indicated that management is a process. A typical management will have five primary functions. These are:

- Planning
- Organising
- Staffing
- Directing
- Coordinating
- Reporting
- Budgeting

According to some writers, the last three are part of the controlling functions.

Thus, the short form 'POSDCORB', coined by Luther H. Gulick and Lyndall Urwick, came into prominence. This represents the initials of the seven functions mentioned above.

- i) **Planning:** We have already spent some time learning about the planning process. However, it is worthwhile to note here that planning is future oriented and determines an organisation's direction. It is a kind of organised foresight as well as corrective hindsight. In other words, planning involves formulation of objectives for the future, working out specific steps or programmes that will contribute to the achievement of those objectives, noting down simultaneously the problems and issues that may have emerged earlier in similar programmes and taking measures to face those problems in an effective manner.
- ii) **Organising:** Organising involves
 - determining activities needed to fulfill the objectives;
 - grouping the activities into manageable units; and
 - assigning such groups of activities to managers. It determines authority and responsibility relationships. These relationships have to be properly coordinated to achieve unity of organisation. It follows, therefore, that the function of organisation consists of:
 - identifying the task that must be performed and categorising them according to time or cost involved;
 - assigning these tasks to the personnel, while defining their authority and responsibility;
 - delegating this authority to their employees;
 - establishing a relationship between authority and responsibility;
 - coordinating these activities.
- iii) **Staffing:** Staffing is the function of hiring and retaining a suitable work-force both at managerial as well as non-managerial levels. This function

is critical since people differ in their intelligence, knowledge, skills, experience, physical condition, age and attitudes. Hence, management must understand the sociological and psychological background of each member of the work-force in addition to their technical and operational competence.

- iv) **Directing:** This function is concerned with leadership, communication, motivation and supervision in order to achieve desired goals. This function is directly concerned with the human factors of an organisation.
- v) **Coordination:** Management activities assigned to different departments of an organisation needs to be coordinated from the point of view of the objectives. Similarly, different activities within a department need to be coordinated. This is necessary for the smooth functioning of an organisation and its departments. Coordination is a part of the regular managerial function, but an organisation may have separate department(s) and procedures for coordination.
- vi) **Reporting:** Management has to keep itself informed about the performances of tasks assigned to departments and sections and to the personnel working at different levels. Reporting refers to information about the progress of tasks performed. The reports must be reliable for effective monitoring. There should be constant flow of performance reports through different levels of organisation reaching upto the highest levels. Management should work out such reporting system for efficient and timely performance of tasks.
- vii) **Budgeting:** Budgeting refers to the allocation of resources for organization's tasks and further appropriating of available resource to different tasks to be performed by the departments/sections. Besides the monetary budgeting, it also becomes necessary to budget the time assigned for the performance of tasks, within the budgeted time and money.
- viii) **Controlling:** It consists of those activities which are undertaken to ensure that the events do not deviate from the planned paths. It involves actions like watching the progress and taking corrective action when desired. The control process is also referred to as monitoring. This process generates information for modification or even creation of new plans. A subsequent unit will engage our attention exclusively for a detailed analysis of this function.

We must conclude our discussion with a brief note on the decision-making process. The decision-making process must be understood as a sequence of events. The sequence of events can be represented as follows:

- Understanding the situation
- Identification of the right problem(s) (diagnosis)
- Search for the analysis of available alternatives
- Selection of the most appropriate solution
- Acceptance of the solution for implementation.

People, area and activities form the setting for rural development for which

management decisions are taken. The understanding of the setting is the prerequisite of management in rural development. Occasionally, inadequate or faulty understanding of the setting leads to wrong management decisions. People who are trained in only management techniques without having the understanding of the rural setting have tried to apply management techniques, but have failed miserably. At the same time, people who have spent a lifetime in acquainting themselves with the setting, but are ignorant of the management process, have found that their sincere efforts do not bring about the desired results.

Check Your Progress II

- Notes:** a) Write your answers in the space provided.
b) Compare your answer with the text.

How are the planning functions different from organising functions?

(*Hint:* Read carefully the text and mark the key words before attempting the answer.)

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Activity I

Suppose you have to manage a dairy project. In what way are the people and the area important for the successful management of a dairy project?

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1.4 IMPLEMENTATION OF RURAL DEVELOPMENT PROJECTS

One of the main functions of management is effective execution or implementation.

Implementation may be regarded as a process by which a set of predetermined activities is carried out in a planned manner with a view to achieve certain established objectives. Implementation is, thus, a subset of a broader set of activities that constitute management. A programme is usually implemented through a series of well defined projects. The scope of a plan is still broader in the sense that it usually comprises a number of programmes. Thus, for its implementation, a plan has to be broken down into a number of programmes, each of which is, in turn, broken down into a number of projects. A project treats the questions, what, who, when, where and how more specifically than does a programme.

Phase of Implementation

Implementation consists of two sub-phases:

- 1) Pre-operation phase
- 2) Operation phase

The pre-operation phase of implementation begins when the feasibility report has been completed and financing has been arranged. The pre-operation phase may be considered to be complete when various components of the project are installed and put into operation.

The objectives of project management in the operation phase are as follows:

- Completion of the project on time;
- Completion of the project within the contemplated cost.

Factors Affecting Implementation

There are several factors which affect the implementation of a programme. A rural manager has to understand their role and function. Let us briefly list them.

- i) **Technical Factors:** Rural projects can be of diverse kinds – irrigation, livestock development, wasteland development, etc. All these involve the use of technology and recruitment of technical manpower, so that all the production functions are effectively tied up to maximize output. Rural development management requires understanding of the technical factors involved and how to manage them.
- ii) **Economic and Financial Factors:** These relate to factors affecting credit, subsidies, pricing, choice of resources, etc.
- iii) **Commercial Factors:** These relate to marketing of outputs, supply of inputs, etc.
- iv) **Socio-Cultural Factors:** Since rural society and economy are closely

interlinked, the project manager has to understand the role of socio-cultural values, tradition, social structure, castes and kinship pattern, social institutions, etc., and social stratification.

- v) **Political Factors:** The power structure, dominant groups, leadership patterns etc., too affect implementation.
- vi) **Institutional, Organisational and Managerial Factors:** The implementation of the project will be affected by the extent to which the organisational and managerial objectives and functions have been clearly laid down, responsibilities clearly demarcated, inter-linkages tied up, and monitoring and project control systems developed.
- vii) **People's Participation:** Involvement of people in the implementation of the project is of key importance. Therefore, the project manager should not only aim at people's participation, but also develop the strategies and the mechanisms.
- viii) **Integration and Coordination:** Since implementation involves several agencies, therefore integration and coordination both vertical and lateral, are absolutely essential.

Activity II

Visit a rural development project in your area. Closely study its functioning and then indicate how different factors are affecting its implementation.

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Delays in Implementation

The most important problem faced in the implementation phase of a project is delay in execution. This is also referred to as slippage of projects, which results in escalation of costs and also the loss of revenue. The result is that the initial assumption in the feasibility report are thrown completely out of gear. The delays are generally caused by internal as well as external factors.

The *internal factors*, which cause delay are:

- Inadequate planning based on wrong and inadequate information – Inadequate financing
- Choice of wrong technology
- Lack of coordination among execution departments
- Absence of delegation of power
- Lack of reasonable norms of accountability
- Poor industrial relations

- Wrong selection of agents
- Non-involvement of people

The **external factors**, which may cause delay are:

- Input problems
- Transportation problems
- Frequent change in administration at the senior level, affecting the continuity of policy
- Lack of public cooperation.

Despite taking all the care, there are bound to be factors beyond the control of a manager, which are likely to cause delay in project implementation. Therefore, it is necessary to build a system, which can correct deviations from the initial expectations and ensure that the schedule of the project is not disturbed. To perform such tasks, there needs to be a scientific system of flow of management information to the executive at various levels in the organisation. We will read about these aspects in the units that follow.

The system of monitoring and control is more effective when, along with the time-frame, there is linkage between the physical work to be performed in each activity and the financial expenditure to be incurred. The operational plan should include the following aspects:

- The breakdown of the project's final objectives into various subsystems and these subsystems into activities and sub-activities.
- The time schedule for starting and completion of each activity and sub-activity and their sequencing.
- Linking of physical work involved in each activity with financial expenditure.
- The timetable indicating the requirements of financial resources during the pre-implementation phase.
- Monitoring and control system.

Activity III

Visit a project in a rural area, which has overrun the scheduled date of completion, and indicate the factors, which have caused delay. You may find factors other than those listed in the text.

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9.5 LET US SUM UP

In this unit, we stated that management is the art of getting things done by directing the efforts of other people so as to bring about the best possible allocation and utilization of human, material and financial resources towards attainment of goals. We also noted the difference between administration and management as perceived by some experts who view administration to be above management. Some experts use the terms interchangeably, while yet others consider management to be a much wider term to include administration.

We identified the main functions of management as planning, organising, staffing, directing and controlling. It was observed that since people, area and activities form the setting for rural development, their understanding is a prerequisite for effective management.

Later in the unit, we discussed the factors (external and internal) affecting implementation of projects and some of the important causes of delay.

9.6 KEY WORDS

- Controlling** : A function which ensures that events do not deviate from the preplanned path, thereby affecting the attainment of objectives.
- Planning** : A function which determines the direction of an organisation.

9.7 SUGGESTED READINGS

Mondy, R. Wayne, Robert E. Holmes and Edwin Flippo (1983), *Management: Concepts and Practices*, 2nd ed., Allyn and Bacon Inc., Boston.

Stoner, James A.F. and Charles Wankel (1988), *Management*, 3rd ed., Prentice Hall of India Private Limited, New Delhi.

Peter F Drucker (1994): *Frontiers of Management* (Heinemann London)

Peter F Drucker (1993): *Managing for future* (Tata Mcgraw Hill)

UNIT 10 PROJECT DIMENSION, IDENTIFICATION AND FORMULATION

Structure

- 10.0 Aims and Objectives
- 10.1 Introduction
- 10.2 Dimension of a Project
- 10.3 Process of Identification
- 10.4 Process of Formulation
- 10.5 Detailed Project Report (DPR)
- 10.6 Let Us Sum Up
- 10.7 Key Words
- 10.8 Suggested Readings

10.0 AIMS AND OBJECTIVES

The basic objectives of this unit are to acquaint you with the following:

- Dimensions of a project;
- Steps involved in project identification;
- Steps involved in project formulation; and
- Format of a detailed project report.

10.1 INTRODUCTION

As discussed in the previous unit, the project life-cycle has many distinct phases. In this unit, we will discuss the first two phases of the project life cycle. However, it will be preceded by a detailed examination of the dimensions of projects including various types of linkages. The discussion will facilitate you in the identification of a number of development projects which will be available for selection and appraisal.

10.2 DIMENSIONS OF A PROJECT

It is generally known that a project would belong to a sector/sub-sector. In the process, it gets identified with a particular sector/sub-sector. In a real-world situation, such identification has been misleading. A project has many more dimensions and it is essential to identify them at the outset.

- i) **Activity group:** The first dimension of the project is the activity group or nature of everyone. The project would belong to any of the following

four activity groups classified according to the nature of the activities. Such activity groups are:

- a) Final commodity-producing sectors,
- b) Productive infrastructural sectors,
- c) Marketing infrastructural sectors, and
- d) Social service infrastructural sectors.

Details about these sectors will be done in a separate unit subsequently. Nonetheless, we will briefly describe their nature.

The nature of this pattern of grouping is such that groups (b), (c) and (d) are complementary to group (a), group (d) stands on its own, as it largely comprises sectors/sub-sectors that provide for minimum social needs. Hence, it leaves little choice of any shift of investment from one sector to another within the group or with (b) and (c).

In terms of implications, consequent to a change or revision in groups, the direction of change needs to be assessed properly. With respect to groups (b) and (c), a revision within-group (a) will indicate the direction in which the revision in these two groups should be done.

For example, let us take milk production. Suppose, there is a revision within group (a) since milk production needs relatively more investment. This would call for, correspondingly, more investment in feeding and veterinary facilities in a group (b) and, similarly, more in marketing infrastructure for animal feed and milk output.

Therefore, a sequential procedure is to be followed while identifying and revising projects.

ii) Independence and complementarity: We have already discussed in detail about this dimension in the preceding units. As discussed, the project may belong to a wider programme in which it may either be independent or complementary to other projects. The attempt should, however, be to avoid conflict of objectives between the projects. This will ensure the maximization of benefits to the beneficiaries and, ultimately, society.

iii) Potential beneficiary group: Since a project is a planned intervention for the development of a certain social segment within a specific time period, the potential segment's interest becomes another dimension of the project. The project would have one or other of the following potential beneficiary groups as targets:

- a) The population of a Community Development Block in general or the inhabitants of a group of villages, e.g.: roads/electrification;
- b) The village community as a whole, e.g., for school, primary health centre, drinking water, public tube well;

- c) Members of a cooperative society, e.g: collection/chilling centre for a milk cooperative, wool carding centre for a weaver's cooperative;
- d) A specific target group, such as an association of target group persons or households for sharing an irrigation tube-well or drinking water well. It may also consist of individual members of households belonging to the target group, e.g., housing, dairy unit, shoe making;
- e) Private individuals, private firms and companies or households in general, e.g., small scale units, dairy units, orchard plantations.

Thus, we can see that the potential beneficiary groups have a wide range and accordingly the dimension of the project varies.

iv) Share in investment (capital structure): The capital structure is determined by the share in investment proposed by the private sector. Therefore, the capital structure is a very important dimension of a project. Perhaps by now, you know why. Depending on the beneficiary group, and their interests, the capital can be obtained either from the public sector, individuals, institutions (private sector), village community or panchayat, cooperative society or a combination of these.

In some extreme cases, investments can be fully public or private. Further, depending on the investors and in what proportions they share the project investment, the capital assets formed under the project can be either shared or owned.

v) Financial interests: Consequent to the capital structure, financial interests in the project could be many. Depending on the contributor's share in the project, the interests would vary. Project finances can come partly through fund transfers among the investors themselves by way of grants and loans or partly in the form of loans from financial institutions. This dimension of the project cannot be overlooked.

vi) Allocation and area specificity: The problem of allocation of time and resources, especially capital, has an important bearing on project identification. Therefore, it calls for proper coordination among different activities.

The spatial distribution and locational structure of economic activities and facilities determine what kind of projects can be formulated and what could be the allocation of different resources. Concepts like growth poles and central places can be used to illustrate the spatial and locational dimensions of the project.

A **growth pole** is a location (a town, village, or area) where one or more major economic activities (e.g., major industries or production centers) are concentrated. A **central place** is a location where a major physical (transport, warehouses, communication) and institutional (marketing, financial infrastructural) facilities are concentrated.

Normally there would be an order of priority amongst growth poles and central places in a region. The order of priority would depend on the extent of concentration of production activities and infrastructural facilities. While some projects may yield better developmental results when located in existing growth poles or central places, other projects may need to be set up to create new growth poles or central places.

Due to the differential effects of the project and the existing facilities, the command areas of growth poles and central places need to be properly assessed before the identification of the project.

Identification and clear specification of all the dimensions mentioned here are necessary prerequisites to proper appraisal and ultimate selection of projects. Formulation of the project blueprint should start with a specification of project dimensions.

Check Your Progress 1

Notes: a) Write your answers in the space provided.

b) Compare your answers with the text.

- 1) List the reasons why the capital structure is an important dimension of a project.

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- 2) Identify the target groups for a family planning programme. Mention which of these you consider to be primary.

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- 3) Indicate two projects in a community development block, which are characterised by complementarity and two in which, there is conflict.

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Activity I Identify some growth poles and central places in your district. Indicate the reasons why you consider them as growth poles and central places.

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10.3 PROCESS OF IDENTIFICATION

The process of project identification starts with the conception of an idea. The ideas have to be tested for their practicality and worthiness. Say in a community development block, the key factor would be the answer to the following questions:

- i) Whether the project implied in the idea would fall within block-level planning activities in terms of scope and investment?
- ii) Whether its technical features, including physical location, or layout alignment. etc. are **prima facie** sound?
- iii) Whether the material conditions of its successful operation in terms of the actual and potential availability of required raw material and energy exist in the block?
- iv) Whether the required skilled manpower would be available either because it already exists or there are conditions for its speedy development?
- v) Whether there is a local demand for the goods or services the potential project is intended to produce. Also if local demand is inadequate then whether the excess output can be sold to urban centres or to neighbouring blocks at competitive prices?
- vi) Whether the particular social minimum needs that the potential project intends to serve is unfulfilled in the block?

If the answer to these questions is an emphatic 'yes', then the project idea should be translated into a project blueprint. And if the answer turns out to be a 'no', the project idea should be dropped or, at best, postponed until favourable conditions for reconsideration arise.

10.4 PROCESS OF FORMULATION

After the series of project ideas have, thus, been identified and tested, individual projects need to be explicitly formulated. The project should be formulated or prepared on the basis of a detailed analysis of specific techno-

economic relationships. Project formulation generally leads to a Detailed Project Report (DPR).

This is the stage at which a detailed study will commence. So at this stage, realistic estimates can be made about how the project can be implemented and of its probable income-generating capacity. Detailed preparation of a blue-print takes time. According to the World Bank, it is expensive, as it may cost 7 to 10 per cent of the total project investment.

The process of formulation or preparation includes all the work necessary to bring the project to a point at which a careful review or appraisal can be undertaken. If the project is selected post the review, implementation of the project can be carried out.

A feasibility study has to be undertaken as the **first step** in project preparation. This step facilitates in deciding whether detailed advanced planning needs to be done. The feasibility study should clearly define the objectives of the project to fit its physical and social environment. This is expected to ensure that the project when implemented will give maximum returns. Details of feasibility analysis will be discussed in subsequent units.

In the preceding Unit, a detailed discussion on the plan, objectives and horizon has already enumerated the steps necessary to frame objectives for the block plan. These objectives will now have to be matched with the physical and social conditions of the village/block as step one.

Since each plan for a project is time-bound, you have to first make an inventory of all the resources in the base year. The accompanying chart will describe the manner or scheme of actions to be initiated by you at the block level, see Figure 1.

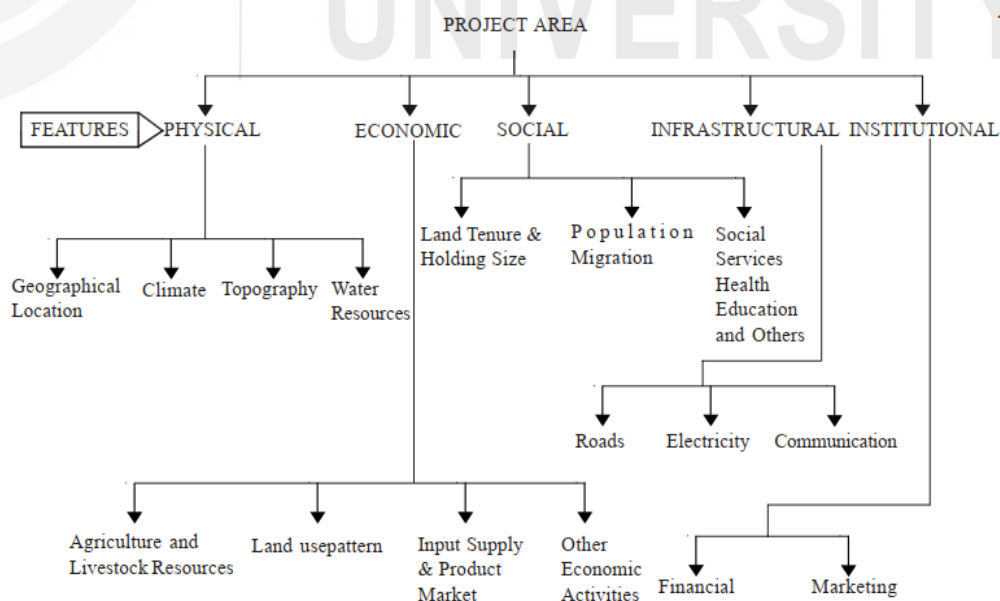


Fig. 1: Details about the features of the project area to be identified while formulating a project

This scheme will describe the existing status of the area where the project will be located and the base from which the project would start. In other words, it will be the framework of **constraints** and **opportunities** for the project.

Check Your Progress 2

Note: a) Write your answers in the space provided.

b) Compare your answer with the text.

- 1) Think of the ways in which you can match the objectives with the physical and social conditions of a block you have visited. (**Hint:** See the chart and the preceding text.)

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10.4.2 Subsequent Step

The next step involves working out the activities and general facilities, their phasing, costs and mode of financing. The organization and management aspects are also worked on at this stage. However, it must be noted that the activities would vary depending on the nature of the project. Therefore, a suggestive outline is given below. The outline can be adapted specially to fit the project which you are formulating.

- i) The Project Description
- ii) Component Details: Works and General Facilities
- iii) Project Phasing
- iv) Cost Estimates
- v) Financing
- vi) Organization and Management
- vii) Production, Markets and Financial Expectations
- viii) Benefits and Justification
- ix) Outstanding Issues
- x) Annexure.

10.5 DETAILED PROJECT REPORT (DPR)

A DPR gives an overall picture of the proposed project and helps understand its prospects. One can very well understand the soundness of his or her proposed project on the basis of DPR and decide whether he or she should go

ahead with the project or not. It is DPR on the basis of which a financing institution or bank appraises the technical feasibility, economic and commercial viability, and financial viability of a proposed project. Based on the appraisal of DPR a FI, bank decides whether to extend financial support to the proposed project or not; and if to be extended what should be the quantum of such assistance

The World Bank recommends that, as a rule of thumb, a DPR should aim at the main text of about twenty-five single-spaced typescript pages for a simple agricultural project and no more than fifty pages for a complex rural development project. This has to be supported by a series of annexures, possibly in a separate volume. As far as possible, the main text should present the project in a form that a non-specialist can understand; specialized backup information – including maps, charts and detailed table-should be reserved for the annexures.

In the beginning, however, it must be noted that the substance of a project's DPR is more crucial than its format. Primarily, because each project will have a different focus and hence different elements of the project will need different emphasis. However, an outline of the format is provided here so that you, as a project formulator, can have a starting point. The checklist of the format, as discussed in previous sections, can now be given as follows:

- i) **Executive Summary:** This section should very briefly give the essential elements of the projects. It will include the rationale, priorities, beneficiaries, main components, investment period, costs, organization, and financial and economic efforts.
- ii) **Introduction:** This section should indicate the origin of the project idea and the agencies responsible for preparing the DPR. This section is basically a non-technical one and should be brief.
- iii) **Background:** This part is mainly for establishing the framework of the project in relation to the environment (natural, economic and social). The material presented here should have a direct bearing with materials to be presented in subsequent sections.
- iv) **Project Rationale:** This section should provide a persuasive discussion based on the process of identification discussed earlier. It should also give the technical, economic and social justification for the selection of the project. Since there are many development strategies all cannot be adopted by the project. Hence an explanation must be provided for choosing a particular strategy. Limitations for the project should also find a place in this section.
- v) **Promoter's Profile:** A promoter should furnish all such personal details in brief in a DPR as would help understand his/her background, personality traits and capability. It should include details such as name and address, his or her age, profile, category (SC/ST/OBC/Minority/Technical/Ex-service personnel/Physically handicapped), sex, educational

qualifications, relevant training received, previous work experience, family occupation, economic status of a family, own financial strength etc.

- vi) **Project Area:** The development opportunities and potential of the project need to be mentioned here.
- vii) **The Project:** This section should define and describe in detail, the objectives of the project and its operationalization. It should include details such as the name of the project, place, manpower required, the investment required, type of ownership etc.
- viii) **Organization and Management:** This section will describe the implementation and monitoring agencies for different aspects of the project. It should also discuss the administrative arrangements, staffing patterns, equipment, functions and powers of the project authority.
- ix) **Production, Markets, and Financial Results:** This section should show why beneficiaries should join the mainstream of the project. The financial and economic feasibility of the project should be provided here.
- x) **Benefits and justification:** This is a crucial part of the DP where all aspects of the project, discussed till now, are to be assessed for social and economic desirability. We will discuss this in the next section for a detailed understanding.
- xi) **Outstanding Issues:** This section is to contain some highlighting issues which the project will have to tackle at some future date.
- xii) **Annexure:** This part contains detailed support for the project. As this section is meant for specialists and experts, it must be prepared with special care. The materials included are statistical tables, maps, drawings, charts, photographs and diagrams. The order of presentation in this section must correspond with the main text presentation.

10.6 LET US SUM UP

By now, we have discussed in greater detail the project life-cycle, viz., Dimension, Identification and Formulation. The status of the project objectives amongst the national plan objectives and the reasons for its selection, the project itself including its technical aspects and costs, and the proposed manner of implementation have all been covered. A special, emphasis on the process of project identification and preparation has demonstrated their crucial role in the detailed project report. This report decides the fate of the project. Hence, in the next Unit, we will emphasize on appraising the project for its economic and financial feasibility.

Activity-3

Take a detailed project report of any project from your own office or from the District Rural Development Agency's office. Study the DPR carefully, list at least five strong and weak points each. Briefly justify your answer.

10.7 KEY WORDS

Capital Structure: Denotes share in investment by various agencies,

Central Place: Location of major physical and institutional infrastructural facilities.

Detailed Project Report (DPR): A report providing details about the proposed report.

Growth Pole: Location of economic activities.

Project Identification: Testing of a conceived idea.

10.8 SUGGESTED READINGS

Gittinger, J. Price, 1982. Economic Analysis of Agricultural Projects (Rev. Ed.2), The John Hopkins University Press: Baltimore and London

Mishra, S.N., 1984, *Rural Development Planning – Design & Method*, Satvahan Publications: New Delhi (essential reading – Chapter 3 and 4)

UNIT 11 PROJECT APPRAISAL: TECHNICAL, ECONOMIC AND FINANCIAL FEASIBILITY

Structure

- 11.1 Aims and Objectives
- 11.2 Introduction
- 11.3 Projects: Meaning and Concept
 - 11.3.1 Project Preparation
- 11.4 Project Appraisal Techniques
- 11.5 Technical Feasibility
- 11.6 Economic Feasibility
 - 11.6.1 Role of Time in Economic Appraisal
- 11.7 Financial Feasibility
 - 11.7.1 Main Characteristics
- 11.8 Key Words
- 11.9 Let Us Sum Up
- 11.10 Suggested Readings

11.1 AIMS AND OBJECTIVES

The present unit aims to orient the learners to the concept of project appraisal and practical aspects of the technical, economic and financial feasibility analysis. In the process, some steps, necessary to identify costs and benefits have been explained.

By the end of this unit, learner should be able to achieve the following objectives:

- Decide why and when to carry out technical, economic, financial feasibility analysis;
- Integrate the three analytical tools, viz. technical, economic and financials, of project appraisal into a sound and feasible detailed project; and
- Gain a good understanding of the different economic measures and the differences between them.

11.2 INTRODUCTION

This unit deals with the concept of project appraisal and various project appraisal techniques. The present unit attempts to provide an understanding of the theoretical foundations of the method of economic and financial

feasibility analysis. There is, therefore, a clear bias in this unit towards providing an answer to the question, “How to carry out an economic feasibility analysis”?

11.3 PROJECTS: MEANING AND CONCEPT

What are Projects?

Projects are the cutting edge of development. Projects are investment activities in which financial resources are *spent* to create capital assets to produce benefits over a certain time.

According to UNIDO project refers to a proposal for an investment to create and develop certain facilities to increase the production of goods and services in a community over a period of time.

The Chartered Management Institute defines a project as “an activity that is carried out to achieve a set quality within a speculated time and cost limits that has a certain start and end. In other words, a project is as an activity for which money is spent in expectation of returns and that logically seems to lend itself to planning financing and implementation as a unit.

11.3.1 Project Preparation

The preparation of a project necessitates consideration of many aspects. The major aspects to be considered in preparation of a project are:

- 1) Technical
- 2) Institutional
- 3) Organizational
- 4) Managerial
- 5) Social
- 6) Commercial
- 7) Financial
- 8) Economic

1) Technical Aspect

It considers the technical feasibility of any project. The technical aspect of a project is concerned with both input supply side and output delivery side. For example, if you want to start an agricultural project in any particular area, you need to examine the soil type, water availability, crops grown, livestock breed suitable for the area, pests prevalent in the area etc. This information can be used in estimating the possible yield and income from agriculture. Such information can be collected through soil surveys, groundwater surveys, collection of hydrological data, primary surveys of farmer household, etc. The information regarding marketing and storage facilities responsible are needed to assess the possibility of marketing and processing of the products.

2) Institutional Aspect

A complete knowledge of the institutional aspect helps in identifying the components of institutional framework that will have a bearing on the project. Some of the elements that constitute the institutional framework include government institutions, project authority, corporate bodies, land tenure systems, banking and credit institutions, religious customs, practices and social mores. There is a need to understand the administrative system of the region where the project has to be undertaken.

3) Organizational Aspects

Here the term organization refers to the structure of the body that would undertake the task of project execution. The proposed organization must have the capacity to carry out the assignments given to it. Some of the basic principles to be followed include:

- Clear lines of authority running from top to bottom of the organization and the chain of command/number of levels of authority should be clear and minimum.
- Clear definition of responsibilities of each authority in writing.
- Decision making power to be placed as near as possible to the scene of action.
- The organization should be kept as simple as possible and should be flexible to adjust to changing conditions.

4) Management

The main task of management is to implement the project objectives within the framework of organizational structure. For good management, a clear definition of functions and activities are required. There is also a need for allocating responsibilities to various agencies for various project activities. A suitable mechanism for coordination of the activities of participating agencies should also be developed. Proper staffing also comes under the purview of the management.

5) Social Aspects

It is very important to assess the social patterns, customs, culture, traditions and habits of the clientele. Various aspects like changes in living standards, material welfare, income distribution, etc. In selecting some projects, weights are assigned for income distribution so that the projects which benefit the lower income group are benefitted. The adverse effect of the project on a particular group is also examined. Preserving the environment and wildlife habitats is given high priority.

6) Commercial Aspects

The commercial aspects of a project involve the arrangements for marketing the output produced by the project and ensuring supply of inputs needed for the project to operate smoothly. There is a need to

assess the effective demand of the project output and the prices that may prevail under the demand and supply situation. The analyst also needs to cautiously evaluate the impact of product supply on the price of the product and the viability of the project under such changed price situation.

7) Financial Aspects

The decisions about undertaking any project, depend on a lot on financial analysis of a project. As there could be many beneficiaries participating agencies of any project, there is a need for separate financial analysis. It involves the project organization to raise the appropriate funds needed to implement the proposed project. A financial feasibility study of a project gives an idea about how much start-up capital is needed, sources of capital, return on investment and other financial considerations.

11.4 PROJECT APPRAISAL TECHNIQUES

Project Appraisal

Project appraisal is regarded as the review of the work plan, estimation of the cost and financing of the project. It also concerns the assessment of the project's organizational and management aspects as well as the viability of the project. Project appraisal criteria can be classified as:

- 1) Non-Discounting Techniques
- 2) Discounting Criteria Techniques

1) Non-Discounting Techniques

i) Urgency

According to this criterion, the projects that are more urgent get preference over those that are less urgent. However, one of the problems involved in using this criterion is to judge the urgency of any project as decision taken may be subject to the personal bias of the decision maker. In view of this limitation, it should not be used for investment decision-making.

ii) Payback Period

It is the length of time required to recover the initial cash outlay on the project. In case cash inflows are constant, then the payback period is calculated by dividing the initial outlay by the annual cash inflow.

On other hand if the cash flow is not constant, e.g. if a project involves a cash outlay of Rs 6,00,000 and generates cash inflow of Rupees one lakh in the first year, one lakh fifty thousand in the second year and one lakh fifty thousand in third year and Rupees two lakh in fourth year, its payback period is four years. As per the

payback period criterion, the shorter the payback period, the more desirable is the project. Firms using this criterion, generally specify the maximum acceptable payback period.

iii) Accounting Rate of Return

The accounting rate of return or the simple rate is the measure of profitability which relates income to investment, both measured in accounting terms. As there are various ways of measuring income and investment, there are a large number of measures for accounting rate of return. The commonly used ones are given:

- 1) $\frac{\text{Average income after tax}}{\text{Initial investment}}$
- 2) $\frac{\text{Average income after tax}}{\text{Average investment}}$
- 3) $\frac{\text{Average income after tax but before interest}}{\text{Initial investment}}$
- 4) $\frac{\text{Average income after tax but before interest}}{\text{Average investment}}$
- 5) $\frac{\text{Average income before interest and taxes}}{\text{Initial investment}}$
- 6) $\frac{\text{Average income before interest and taxes}}{\text{Average investment}}$

Decision-making: The higher the accounting rate of return, the better the project.

Evaluation:

- It is simple to calculate.
- It is based on readily available accounting information.
- It considers benefits over the entire life of the project.
- Though the income data of the entire life of the project is required, one can work out accounting rate of return even if the complete income data is not available by taking income from a typical year.

Disadvantages of accounting rate of return:

- It ignores the time value of money.
- Due to numerous measures of accounting rate of return it may create confusion.

iv) Debt Service Coverage Ratio

The debt service coverage ratio is generally used to find the financial worthiness of the projects which needs long term financing. The formula is

[net profit + interest (on long term loan) + depreciation] / {interest (on long term loan) + principal loan}.

Decision-Making: Generally, the financial institutions regard a debt service coverage ratio of 2 as satisfactory.

2) Discounting Techniques

i) NPV (NET PRESENT VALUE)

It is the difference in between the present value of cash inflows and the present value of cash outflows. NPV is used to analyze the profitability of an investment/ project in capital budgeting.

The NPV analysis is sensitive to the reliability of future cash inflows that an investment/project will yield.

The formula for **NPV** is:

$$NPV = \sum_{t=0}^n \frac{(\text{Benefits} - \text{Costs})_t}{(1+r)^t}$$

where:

r = discount rate

t = year

n = analytic horizon (in years)

Decision-making:

The acceptability of a prospective project depends on the NPV. If the NPV is positive prospective project should be accepted. If the NPV is negative, the project should probably be rejected because cash flows will also be negative.

Features of NPV:

- The NPV assumes that the intermediate cash inflows of the project are re-invested at a rate of return equal to the firm's cost of capital.
- The NPV of a simple project decreases as the discount rate increases, the decrease in NPV however is at a decreasing rate.

Merits of NPV:

- It considers the time value of money.
- It considers the cash flow stream in its entirety.

The NPV's of various projects can be added. The NPV of a scheme consisting of two projects A and B will simply be the sum of NPV's of these projects individually.

$$NPV (A+B) = NPV(A) + NPV(B).$$

Let us understand the net present value (NPV) for following cash flow stream:

Year	Cash Flow
0	-10,00,000
1	2,00,000
2	2,00,000
3	3,00,000
4	3,00,000
5	3,50,000

The cost of capital k for the firm is 10 per cent. The net present value of the proposal is:

$$NPV = \frac{10,00,000}{(1.10)^0} + \frac{2,00,000}{(1.10)^1} + \frac{2,00,000}{(1.10)^2} + \frac{3,00,000}{(1.10)^3} + \frac{3,00,000}{(1.10)^4} + \frac{3,50,000}{(1.10)^5}$$

$$= -5273$$

Since the decision rule associated with the net present value is to accept the project if the NPV is positive and reject if it is negative. Here, the decision should be to reject the project as the NPV is negative.

i) Internal Rate of Return (IRR)

In general, any project having the highest internal rate of return is supposed to be more desirable and would probably be considered the best and undertaken first. Internal Rate of Return (IRR) is also referred to as “economic rate of return (ERR)”. It is the Project Appraisal discounted rate in the equation:

$$0 = \frac{CF_0}{(1+r)^0} + \frac{CF_1}{(1+r)^1} + \dots + \frac{CF_n}{(1+r)^n} = \sum \frac{CF_t}{(1+r)^t}$$

CF_t = cash flow at the end of the year t

r = discount rate

n = life of the project

In the internal rate of return, we set the net present value equal to zero and determine the discount rate which would also be the internal rate of return.

E.g. Consider the cash flow of a project

Year	Cash Flow
0	-1,00,000
1	30,000
2	30,000
3	40,000
4	45,000

The internal rate of return is the value of r which satisfies the following condition.

$$1,00,000 = \frac{30,000}{(1+r)} + \frac{30,000}{(1+r)^2} + \frac{40,000}{(1+r)^3} + \frac{45,000}{(1+r)^4}$$

The calculations of r consist of a process of trial and error. We try different values of ' r ' till we find that the right-hand side of the above equation is equal to the left-hand side. By putting the value of ' r ' as 12 we get 1,07,773 for 14 it is 1,03,046, for 15 it is 1,00,802 and for 16 it is 98,641. Since at 16 per cent, the value is less than 1,00,000, we conclude that the value of ' r ' lies between 15 % and 16%.

At 1 per cent difference (between 15 and 16 per cent) corresponds to a difference of 2161 (1,00,802 - 98,641). The difference between the net present value at 15% (1,00,802) and that at present target value (1,00,000) is (1,00,802 - 1,00,000) is Rs.802. This difference will correspond to a percentage difference of $802/2161 = 0.37$.

Adding this number to 15 per cent we get the value as 15.37 per cent. However, the actual rate of return that a given project ends up generating often differs from its estimated Internal rate of return. A project with a higher IRR value than other available options would provide a much better chance of strong growth.

ii) Benefit-Cost Ratio (BCR)

A Benefit Cost Ratio is an indicator, used in the formal discipline of cost-benefit analysis to evaluate the overall value for money of a project. It is the ratio of the benefits of a project proposal, expressed in monetary and non-monetary terms relative to its costs.

The benefit-cost ratio (BCR) represents the ratio of total benefits over total costs, both discounted as appropriate. The formula for calculating BCR is:

Year	Benefits
Year 1	25,000
Year 2	40,000
Year 3	40,000
Year 4	50,000

he benefit cost ratio of this project will be

$$CR = \frac{25,000}{(1.12)} + \frac{40,000}{(1.12)^2} + \frac{40,000}{(1.12)^3} + \frac{50,000}{(1.12)^4}$$

$$= 1.145$$

Decision making: If BCR is >1 , the project should be accepted and would be beneficial.

BCR $=1$, we interpret it as being indifferent.

BCR <1 , the project should be rejected.

The BC ratio is preferable to NPV as this criterion measures per rupee of outlay and it can discriminate between large and small investments.

Check Your Progress 1

- Note:** a) Use the space provided for your answer.
b) Check your answer with the above text.

1) What are projects? How do they help in development?

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2) Describe with an example the technical aspects to be considered while preparing a project?

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11.5 TECHNICAL FEASIBILITY

Technical Feasibility analysis is aimed at ensuring that all the physical facilities needed are available and the best of the alternative facilities are chosen for successful implementation of the project. It includes the following :

- 1) Selection of proper location of the enterprise,
- 2) Selection of appropriate Technology, Plant & Machinery,
- 3) Availability of Raw Materials, Resources and other Inputs,
- 4) Establishing the Schedule of Project Implementation.

Selection of proper location of the enterprise is a pre-requisite for success of the project. The word location refers to the general location of the enterprise, whereas the word site refers to a specific area within the location. All types of projects have their own requirements but some of the important factors that must be kept in consideration while selecting the location are:

- 1) The availability of land, the cost of land and presence of connecting roads.
- 2) Their proximity to the markets
- 3) The availability of transportation and communication facilities.

To ensure efficient and economical operations, selection of appropriate technology, plant and machinery is essential. The decisions must be taken with careful consideration of demand and fluctuations in demand. For successful implementation of the project, availability of raw materials and other inputs is also important. The various input requirements like raw materials and labour must be estimated on the basis of estimated turn over. The availability of inputs in the right quantity and quality on a regular and continuous basis is essential

for the continuity of operations. If necessary, the enterprise may also create its own sources of supply. Any delay in the implementation of project quite often creates cost over-runs and may affect the profitability of the enterprise. In case of large and complex projects, advanced scheduling techniques like programme evaluation and review techniques are utilized.

The technical feasibility analysis of the project should also take into account the systems for safe and non-polluting disposal of effluents. Use of non-conventional/renewable sources of energy like energy from sun, windmills etc. must also be taken into account.

Check Your Progress 2

Note:a) Use the space provided for your answer.

b) Check your answer with the above text.

- 1) Bring out the importance of technical feasibility analysis in ensuring success of the project.

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11.6 ECONOMIC FEASIBILITY

Economic feasibility analysis is a quantitative measurement of economic efficiency of feasible project variants. This analysis is carried out so that one of the variants could be chosen on the strength of its superiority over the others. Economic analysis serves to quantify differences between alternatives and reduce them to a basis, which enables project comparison. The importance of the use of these methods, however, varies with the alternatives under consideration. Thus, project economic analysis involves techniques for comparing and deciding between alternatives on the basis of monetary or economic desirability. The use of these techniques is very important, for there is much to be gained or lost by virtue of the particular alternative chosen. Let us, therefore, examine these concepts to develop an answer to the question: How the economic analysis should be carried out?

Economic Efficiency

The central and eventual purpose of economic analysis of project proposals is to work out on efficiency measures. Obviously, the question which arises here is: what are economic efficiency measures? Economic efficiency is derived from the primary categories of inputs and outputs of the project. A derivation becomes possible only when input and output quantities have been converted into their respective values by multiplying them with respective prices or unit values.

A necessary and indispensable condition for derivation of economic efficiency, therefore, is that the price or unit value of inputs, and outputs must be given. Being a derived value concept, an economic efficiency measure represents or stands for a relation. At this point you must ask the question: What is this relation? Or alternatively, how is this relation expressed? Fundamentally, this relation can be expressed in two possible ways:

- a) As a difference of value of sums of inputs and outputs, and
- b) As a ratio of output to input value sums.

While the first gives the absolute measure, the second gives the relative measure of a project's efficiency.

- i) **Differences in Efficiency Measures (Private and Social):** Different economic agents value the same things differently because their goals, objectives or motivations are usually different. Take for instance a dairy project where one of the projects is straw for feed. A peasant who participates in the project may have to buy a part of his straw requirement from other peasants at the prevailing market price of straw in or around his village. The value of this input for this individual peasant is the quantity of straw purchased multiplied by the price paid. Consider now the value of the straw input to the project from the angle of all the peasants, that is, from the angle of the society as a whole. Straw is a by-product of agriculture and it has hardly any other use. Therefore, for society, the value of the straw input to the project is virtually nil. The difference in the value of straw input alone will be different in the two cases. While in the first case, the estimate will give a measure of the project's private economic efficiency, in the second case, it will give its social economic efficiency.

This type of difference occurs irrespective of whether the proposed investment in the project is wholly private, public or any shade between the two. All these types of project investments occur in our country because of the mixed economy with various forms of property ownership.

- ii) **Public and Social Economic Efficiency Measures:** There is also a difference between public and social economic efficiency. The public is identical with the state and with that of the agencies of the state. But the state is not identical with society; therefore, the public and social are not identical. To further clarify the difference, let us think of the proposed investment in the dairy project as wholly public and let the project be run and managed by a public agency. The goal of the public agency is to produce and supply milk at competitive prices to a city population, otherwise served by private suppliers. Straw is once again an input to the project.

The management, like the individual peasant, has to buy its straw input requirement from the nearby peasants at the prevailing market price in the area or buy it elsewhere and transport it to the animal shed in the project area. The value of this straw for the public management agency of the project, similar to that of the individual peasant, is the quantity purchased multiplied by the price paid.

But, for society, for the reason given earlier, the value of the straw input to this wholly public project remains nil. Therefore, for this difference in the value of the straw input alone, other things being the same, the public economic efficiency measure of the project would be quite different from its social economic efficiency measure.

We can conclude that for any project, there are different measures of its economic efficiency. These relate to the economic agents or institutions involved in the project, be they private, public or any other. The same conclusion applies if output values are considered, since different economic agents and institutions may value the same project output differently.

Costs and Benefits

Costs and benefits of a project are derived from its input and output values. Differences in input and output values, according to the agents and the institutions involved, naturally lead to differences in the private, public and social costs and benefits of a project. Therefore, a proper identification of the costs and benefits stream becomes the primary task for any project formulator.

At first it might appear that no matter who incurs them, input values together constitute project costs. Similarly, no matter to whom these accrue, all output values together constitute project benefits. There are, however, two factors, which make a difference. **First**, from the social angle, a negative output value, the value of socially undesirable or harmful product constitutes a social cost of the project. A negative input value, therefore, constitutes a social benefit.

Second, from the private and public angle, besides input costs, the project may involve incurring of non-input costs on the one hand and receiving of non-output benefits on the other. Input costs are traditionally described as source costs to the material transformation process within the scope of the project.

Let us examine the concept of non-input costs and non-output benefits for easy identification.

- i) **Non-input Private or Public Costs:** The non-input private or public costs are of two types: a) Taxes and Duties b) Sales Promotion Charges

There is, however, much debate about the treatment of depreciation and interest charges on the instalments of loans, all of which give the appearance of being non-input costs from the private or public angles.

Depreciation is a financial accounting mechanism employed by private and public industrial firms to show how much physical capital assets have been used during an accounting period.

The inclusion of depreciation charges in the private or public cost would, therefore, mean double counting of the input cost of the asset in point. Hence, depreciation is irrelevant for assessing a project's economic cost, public or social. But depreciation is very relevant for assessing the financial viability of a project, as depreciation charges are available to management as a source of funds.

Similar is the case of repayment of loan instalments. Since the cost of an asset has been incurred when acquiring it, showing repayment as costs would again mean double counting. Once again, however, repayment of instalments is relevant for analysing the financial viability of the project.

It must be noted that in the financial balance-sheet or in the cash flow statement of the project, while repayments of instalments are shown as expenditure, the loan amount outstanding is shown as receipts, thus avoiding double counting. In economic accounting, the asset cost simply takes the place of the equivalent loan. Interest is truly a non-input cost, a payment for ownership rights of investible funds. Those who save and advance loans for investment in private or public projects must be compensated for the sacrifice of consumption, while they advance funds. But one of the central concerns for making a project economic analysis from the private or public angle is primarily to check whether the project would generate sufficient returns to be able to pay interest on the capital borrowed. Precisely for this reason, interest has to be kept out of the private or public economic cost of the projects. Adopting this practice, the project evaluator should be able to show that the project's private or public rate of return is greater than the relevant rate of interest. This would simply mean that the project is worthy of being undertaken.

- ii) **Non-output Private or Public Benefits:** Some of the projects have non-output benefits from the private or public angles. These may be in the form of rebates and exemption from states taxes and duties of various kinds, subsidy on sales and procurement or purchase of output at a support price. Perhaps you can now visualize the group of nonoutput benefits accruing to a private or public agency. You must also be able to understand the effect of such concession on the project output.

Well, the effect is an increase in the net value of project output compared with the net value. This takes place only if the net output value had been assessed at the open market price inclusive of taxes and without the subsidy and support through state purchases. In brief, the private or public total cost of the project, thus, comprises direct input and non-input costs incurred by the private or public agency concerned. The total benefits, on the other hand, comprise direct output and non-output benefits received.

The net of total benefit over total cost from the private or public angles gives us the net revenue, income, earnings, profit, surplus, etc. The general rule, therefore, for getting at respective costs and benefits of the project is: Whatever is to be paid by way of input and non-input expenditures would go as costs and whatever is to be received by way of output and nonoutput revenues would be identified as benefits. These would be applicable to the private or public agency goals.

- iii) **Social Costs and Benefits:** In regard to social costs and benefits of the project, you may face a situation that is simultaneously simpler and more complicated.

The process of identification is simpler because our attention is now confined to input and output related costs and benefits alone. There are no non-input costs and non-output benefits of the project from the society's angle.

The complication arises because input costs and output benefits have to be traced both directly and indirectly. It not only has to be directly incurred or generated by the project, but also indirectly, elsewhere in the economy, by other projects, as a consequence of this project's presence. These are generally described as external effects, which may be external economies or diseconomies.

11.6.1 Role of Time in Economic Appraisal

Time is an essential feature of almost every aspect of a project. As we have seen in a preceding unit, every project has an economic life starting with the first year of construction and extending up to the year in which it is scrapped. At the time of preparing its blue-print and its appraisal, its life lies wholly in the future. This means that project inputs, outputs and also direct or indirect costs and benefits are all future dated.

The project profile is accordingly a future time sequence of inputs and outputs, of costs and benefits running over its life. The project profile consequences, therefore, must be properly assessed.

Present and Future Economic Values

This consequence pertains to costs, benefits or their components, even of same amounts, but occurring at different dates because they do not have the same present value. Let us explain this concept of present and future values with some illustrations.

- i) **Present value:** The present value series are independent of time and, therefore, ordinary arithmetical operations can be performed with ease. This being the case, the sum of the present value series of costs and benefits gives us the total present value of the project's costs and benefits. The difference between the benefit and cost sum gives us the present value of net benefits or net present value of benefits. How does

one obtain a present value series of costs and benefits when the project life lies in the future?

- ii) **Future value:** Just as the saying goes that a bird in hand is better than two in the bush, a rupee's worth of benefit received five years hence (from now into future) has far less value than if received now. The farther away in the future a rupee of benefit lies, the lower is its present value. The position is the same with respect to cost, which is actually negative benefit. And it happens because all of us – individuals, institutions and even society as a whole – discount the future *vis a vis* the present.

In other words, we attach less and less weight (importance) to a unit of benefit or cost the farther and farther it occurs in the future, and the rate at which these weights decline is called the rate of discount. Private individuals with a limited span of life discount the future at a rate higher than that at which society or government would do. Such a difference in discounting of future gives rise to different private and social rate of discount. The future time sequence of project costs, benefits and their components, whether private, public or social can, therefore, be converted into a corresponding present value series, provided we are given the relevant rate of discount.

Now, the most important question in the process of project appraisal revolves around the discount rate. One has, therefore, to decide as to what rate of discount should be utilized. So far as private and public discount rates are concerned, the rates of interest at which the public project agency, private individuals, households or firms would borrow or lend investible funds, in part or whole, provide the appropriate substitutes for these rates of discount.

Check Your Progress 3

- Note:** a) Use the space provided for your answer.
b) Compare your answer with the text.

- 1) Can you think of differences in efficiency measures? If yes, explain why?

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11.7 FINANCIAL FEASIBILITY

In a simple way, financial feasibility or viability may be defined as the capacity of a project to meet or pay for its cash obligations or liabilities. When do the cash obligations arise? The cash payment obligations arise when

private or public economic agents involved in the project borrow funds for meeting the whole or part of their investment and operating costs.

If an agent meets costs wholly out of its own funds, the question of appraising the financial viability of the project from the angle of this particular agent simply does not arise, since its economic efficiency measures would alone suffice to guide a decision.

It has been observed that generally an economic agent intends or decides to borrow funds only when he hopes to make enough earnings out of the projects. The agent should be able to pay back the borrowed funds on creditor's terms. Therefore, it should be clear to you that the problem of financial viability arises only when the project is expected to generate cash incomes in the hands of the private or public agents. And, these agents depend upon borrowings to meet their investment and/or operating costs. To further elucidate the point, consider a promotional or infrastructural service project where private agents make complementary investment. There is a loan provision as well for the private agents before they can reap the benefits of the project. Here, it becomes necessary to appraise the project's financial viability from the angle of these private agents. For example, consider an Integrated Cattle Development Project operating in a rural area. The public sector, under this project, is to provide artificial insemination (AI) and health cover either free or at a nominal charge. There is a provision of financial assistance in the form of a bank loan to the rural households of the area for the purchase of a good quality she-buffalo or cow. The main aim of the project is to enhance milk production in the area. The public sector, by way of incentive, has the provision for a small amount of subsidy on this investment. There is no point in appraising the financial viability of a project like this from the public angle, whereas such an appraisal is extremely necessary from the angle of private households.

On this basis, there would be any number of plan projects financed out of public sector resources that do not need a financial viability appraisal from the public angle. This is so, primarily because these are not revenue earning projects, e.g., projects especially meant to meet the social consumption needs (Minimum Needs Programme projects). Promotional and extension service projects would not call for a financial viability appraisal from the public sector's viewpoint.

11.7.1 Main Characteristics

The foregoing discussion has implicitly mentioned various features of the financial viability techniques for appraising a project. We will, however, explicitly discuss three main characteristics of the financial viability analysis, since our discussion on indicators solely depends on its clear identification and understanding.

Commercial Angle

A project, which poses financial viability problems and demands an appraisal, necessarily has a purely commercial focus or angle. This is because a public or private agency, which borrows funds to purchase project inputs must sell the project output and earn cash income to be able to pay back the borrowed funds along with the interest. A commercial success or profitability is, thus, an essential precondition of a project's financial viability.

Partial Commercial Angle

A project depending upon borrowing need not, however, be 100 per cent commercial in the sense that all its inputs are purchased and all of the outputs sold. In our earlier illustration of the cattle development project, a rural family may buy a cow or a she-buffalo out of the loan advances, but the labour of upkeep and part of the animal's feed may be the family's own. On the other hand, the family will seldom sell all the milk produced. A part is invariably retained for self-consumption even if it is only for the infants and the sick. The larger the part retained in this manner, the smaller will be the cash income received and lower the financial viability. This does not mean that the part retained for self-consumption is not a benefit of the project.

The economic worthiness of the project, inclusive of retained benefit, may be high from both the private family and social angles. However, it may, on the other hand, turn out to be a commercial failure and financially an unviable proposition. It becomes clear, therefore, that a loan-based project ought to be appraised for financial viability even when it has been judged worthwhile by the economic efficiency criteria from private, public and social angles.

Cashflow Statement

You may be aware that any project involving commercial transaction with borrowed funds in different proportions will have a flow of cash. Therefore, a cash flow statement tells you about the health of the project. It is a crucial characteristic. Perhaps you may be tempted to ask the question, why is it a crucial characteristic?

The answer is not difficult to seek. The cash account of the project in question is the main instrument for appraisal of financial viability. There would be at least two cash accounts of the project, mainly because of the commercial nature of the transaction involving loan funds. The first may be for the public sector authority and the second for the typical or representative private household. Each account will tell us about the financial viability of the project from the angle of the two agencies concerned.

Therefore, a year-wise cash account running over the lifetime of the project is called a cash flow statement. By its very nature, this statement is a forecast of likely cash receipts and cash payments of the agency concerned for each year of the project's life.

- i) **General guiding principle:** The general guiding principle for preparing a cashflow statement is that of inflow and outflow of cash, no matter on which account an inflow or outflow actually occurs. For example, a rupee of cash received or retained, whether out of own funds, borrowing(s), subsidy, sale of output or of an input like an old machine, is an inflow. Similarly, a rupee of cash paid, whether as repayment of loan, instalment or interest for the purchase of project inputs, by way of taxes and levies, or as loan advances to third parties, becomes an outflow. It is important that the cashflow statement should be made as comprehensive and detailed as the project team can possibly make it. However, it should be ensured that the forecast of each item of inflow and outflow has been made reliably.
- ii) **Purpose:** A cashflow statement serves two purposes: First, it tells us what the annual financial position of the agency concerned will be, based on the net inflow. The net inflow is the cash in hand and is obtained on deducting the outflows from the inflows. The net inflow can also be used to initiate certain remedial measures. For example, revision of the schedule of loan repayment, making arrangement of a larger level of working capital in some years and so on. The net inflow is most likely to be negative in the initial years of a project's life. It could be because there may be more outflow than inflows in the period of construction. Once, however, the project has begun to yield cash income, the expected positive net inflow may be indicative of the financial soundness of the project. Such a position also indicates that loan liability is also being regularly met.

Second, by choosing relevant items from the inflow and outflow segments of the statement, one can work out indicators of financial health in the form of ratios and rates. A number of such ratios and rates are computed by industrial firms to assess the health of the firm from different angles. The rural development project authority, however, should compute a few key indicators of financial viability during the appraisal stage.

Check Your Progress 4

Note: a) Use the space provided below.

b) Compare your answer with the text.

- 1) What purpose does a cashflow statement serve?

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11.8 KEY WORDS

Average return on: Average yield or aggregate benefits produced by an investment.

Benefit-Cost Ratio: The ratio of present value of costs to present value of benefits.

Cash flow statement: A cash account showing the likely cash receipts and payments for each year of the project.

Depreciation: Loss of value in an asset due to use, wear and tear, ageing, etc.

Discount Rate: The rate at which a reduction is made in the future value.

Economic Feasibility: A quantitative measurement of economic efficiency of feasible project variant.

Financial feasibility: The capacity of a project to meet or pay for its cash obligations or liabilities.

Financial rate of return: A rate of discount derived internally from the cash flow statement at which the NPV of the investment is equal to zero.

Internal Rate of Return: The rate of discount at which the NPV (Net Present Value) equals zero.

Marginal: Pertaining to the last additional unit, positive or negative.

Payback period for: A period in which the initial investment on replacement period of the project is recovered.

Sensitivity: Degree of showing the effect of positive or negative change.

Shadow Prices: It is the price, which indicates the real resource value of inputs and/or outputs.

11.9 LET US SUM UP

Projects are the cutting edge of development. In this unit, we have read about the meaning of projects. The unit also discusses the various project appraisal techniques, technical, economic and financial feasibility. Technical Feasibility analysis is aimed at ensuring that all the physical facilities required are the best of the alternative facilities, selected for the successful implementation of the project. The financial feasibility analysis is carried out only when the project is expected to generate cash incomes whether in the hands of private or public agents.

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UNIT 12 PROGRAMME IMPLEMENTATION-MONITORING AND EVALUATION

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12.0 AIMS AND OBJECTIVES

This unit aims to explain the concepts and processes of implementation, monitoring and evaluation of rural development projects. It offers a scope for use of management techniques in rural development projects. On completion of the unit, you should be able to:

- Understand concepts such as implementation, monitoring and evaluation;
- Delineate phases of implementation;
- Identify factors affecting implementation of projects;
- Know reasons for delay in implementation of projects;
- Lay down the process of developing a good monitoring information system;
- Carry out revision and updating of the monitoring time schedule;

- State the objectives of evaluation;
- Find out main dimensions of evaluation; and
- Point out the techniques and criteria for evaluation.

12.1 INTRODUCTION

No matter how systematically a project is planned, there are often found deviations between what is intended and what is actually achieved. The deviations may be in terms of time schedule or resource use or flow of benefits from the project. Most commonly identified cause for such deviations is bad management. Whatever be the degree of accuracy with which project execution plan is prepared, everything rarely goes exactly according to it. There may be changes, deviations and delays in the implementation. These need to be constantly and continuously monitored, so that time and cost overruns can be minimized and appropriate decisions can be taken. Similarly, evaluation is also an important function of modern management system, finding about effectiveness of the programme in achieving the stated objectives. Evaluation also necessitates assessing the time and effort involved in reaching the aim. Notably, evaluation is different from appraisal with respect to the time when it is carried out. An appraisal is an *ex-ante* evaluation when the project details are estimated for the future whereas evaluation is basically an *ex-post* function when the project has been implemented and events/activities concerned have occurred. This unit highlights measures for better management of rural development programme through proper implementation, monitoring and evaluation.

12.2 IMPLEMENTATION OF RURAL DEVELOPMENT PROJECTS

One of the main functions of management is effective execution/implementation. Implementation is a process of carrying out a set of predetermined activities in a planned manner with a view to achieve certain stated objectives. Implementation is, thus, a subset of a broader set of activities that constitute management. A programme is usually implemented through a series of well-defined projects. Scope of a plan is still broader in the sense that it usually comprises a number of programs. Thus, for implementation, a plan is broken down into a number of programmes and each programme, in turn, is broken down into a number of projects. A project treats questions like what, who, when, where and how more specifically than does a programme.

12.2.1 Phases of Implementation

Implementation consists of two sub-phases:

- 1) Pre-operation phase
- 2) Operation phase

The pre-operation phase of project implementation begins when the feasibility report has been completed and financing has been arranged. This phase is complete when various components of the project are installed and put into operation.

The operation phase of implementation has the following objectives:

- Completion of project on time
- Completion of project within the contemplated cost.

12.2.2 Factors Affecting Implementation

As several factors affect implementation of a rural project an implementer has to understand their interventions. Let us briefly explain them.

- i) **Technical Factors:** Rural projects can be of diverse kinds – irrigation, livestock development, wasteland development, etc. They involve the use of technology and recruitment of technical manpower to effectively combine all the production functions so as to maximize output. Implementation of rural development projects requires understanding of the technical factors involved and how to manage them.
- ii) **Economic and Financial Factors:** These factors affect credit, subsidies, pricing, choice of resources, etc.
- iii) **Commercial Factors:** These factors relate to marketing of outputs, supply of inputs, etc.
- iv) **Socio-cultural Factors:** Since rural society and economy are closely interlinked, the project manager has to understand the role of socio-cultural values, tradition, social structure, castes and kinship pattern, social institutions etc. and social stratification.
- v) **Political Factors:** The factors like power structure, dominant groups, leadership patterns etc. too affect implementation.
- vi) **Institutional, Organizational and Managerial Factors:** Implementation of project is affected by the extent to which the organizational managerial objectives and functions have been clearly laid down, responsibilities clearly demarcated, inter-linkages tied up and monitoring and project control systems developed.
- vii) **People's Participation:** Involvement of people in project implementation is of key importance. Therefore, the project manager should not only aim at people's participation, but also develop the strategies and the mechanisms.
- viii) **Integration and Coordination:** Since implementation involves several agencies, integration and coordination, both vertical and lateral, is absolutely essential.

12.2.3 Delays in Implementation

The most important problem faced in the implementation phase of a project is delay in execution which is also called slippage of project. This causes escalation of costs and also loss of revenue and as a result the initial assumption in the feasibility report is completely thrown out of gear. The delays are generally caused by internal and/or external factors.

The *internal factors* that cause delay include:

- Inadequate planning based on wrong and inadequate information
- Inadequate financing
- Choice of wrong technology
- Lack of coordination among execution departments
- Absence of delegation of power
- Lack of reasonable norms of accountability
- Poor industrial relations
- Wrong selection of agents
- Non-involvement of people.

The *external factors* which may cause delay are:

- Input problems
- Transportation problems
- Frequent change in administration at the senior level, affecting the continuity of policy
- Lack of public cooperation.

In spite of all care taken, there may be factors, beyond the control of a manager, causing delay in project implementation. Therefore, it is necessary to build a system which can correct deviations from the initial expectations and ensure no disturbance in the project schedule. It is to be a scientific system of flow of management information to be executed at various levels in the organization. The system of monitoring and control is more effective when, along with the time-frame, there is linkage between the physical work to be performed in each activity and the financial expenditure to be incurred. An operational plan includes the following aspects:

- Breakdown of the project objectives into various subsystems and these subsystems into activities and sub-activities
- The time schedule for starting and completion of each activity and sub-activity and their sequencing
- Linking of physical work involved in each activity with financial expenditure
- A timetable indicating the requirements of financial resources during the pre-implementation phase
- Monitoring and control system.

You should visit and closely study its functioning to identify different factors, even those enlisted in the text, that affect its implementation.

Check Your Progress 1

Note: a) Use the space given below.

b) Compare your answer with the text.

1) Discuss phases in implementation of rural development projects.

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2) Explain different factors that affect implementation of rural development projects.

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12.3 MONITORING OF RURAL DEVELOPMENT PROJECTS

You must have come across the word ‘monitoring’ earlier. Let us understand what monitoring is. Monitoring in project implementation is a device of overseeing its operations through information feedback to ascertain whether:

- The activities are in the desired direction so as to enable realization of the objectives;
- The activities are proceeding as per the given time schedule/ahead of schedule/behind schedule;
- The target population is being reached;
- The achievement of targets and the benefits expected to be generated are being realized;
- Coordination between various departments involved in the project execution is taking place; and
- Participation of the community (where envisaged) is sufficiently strong to sustain the efforts made through the project.

Thus, monitoring is basically an aid to managerial decision-making by

providing immediate information feedback. This helps in detection of problems and, also, in taking timely corrective actions in relation to areas which are lagging behind. It, thus, provides a critical appreciation of the current status of the project.

Let us illustrate the above points. Suppose that a poultry complex is under construction. According to the original schedule for project implementation, a survey to identify beneficiaries in the village is to be completed within two months from the date the project has started. While monitoring the progress of implementation after three months, the Block Development Officer came to know that the survey is not yet over and is likely to delay the overall development project. Under such circumstances, the decision of the officer could be to provide more manpower to accelerate the survey activity, if this is the only reason for the delay.

Similarly, the project could have allocated financial resources for providing subsidy to families who would take up poultry, piggery, dairy development and some other activities for a livelihood. After three months of project execution when the officer monitors the progress, he may come to know that many families prefer poultry farming, but there is no demand for piggery in that area. With such information, the officer may decide to reallocate the resources from piggery project to poultry farming.

The officer may come to know that many families who have gone for dairy development activity are not able to sell milk due to inadequate facilities for collection of milk production or have suffered for want of fodder for milch animals. Using the information provided in the monitoring report, the Block Development Officer may arrange for creation of milk collection centers and supply of feed and fodder.

Hence, a proper monitoring of the project activities and events helps in deciding whether:

- The project schedule can be retained in its original form or it needs revision to complete the project within the stipulated time;
- Any resource reallocation is to be made within the project so as to achieve the overall objectives; and
- Any additional support or infrastructure is to be created to sustain the efforts made.

12.3.1 Information System

The above illustration makes it clear that for any mid-course corrections or managerial decisions, the Project Manager requires information. However, mere compilation of information is not enough. For the purpose of monitoring, the information should be analyzed and the analysis should lead to conclusions on the project status. The conclusions, thus, arrived at should be used for decision-making. The decision has to be implemented and later

followed up to examine whether the decisions improve the situation and the rate of progress. Such a system of obtaining information, reviewing information and taking corrective measures is the essence of project monitoring. This is explained through the following diagram:

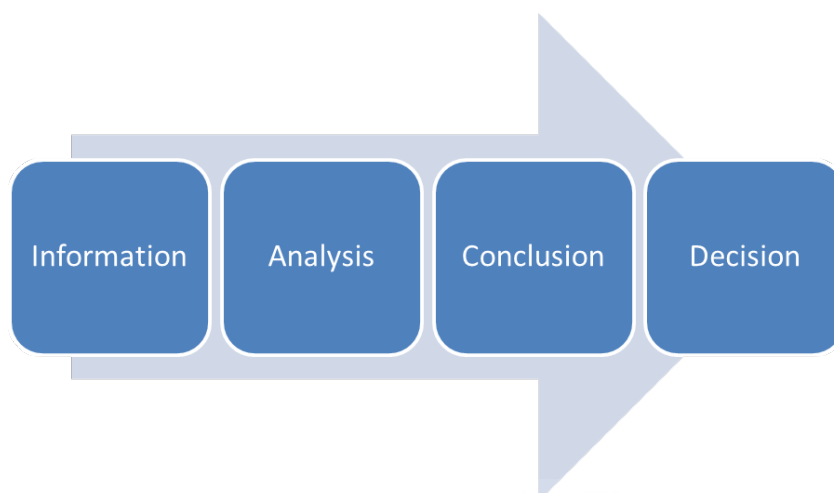


Fig. 1: Process of Project Monitoring

The information required for project monitoring should satisfy the following criteria:

- The information should be reliable;
- It should be relevant to the particular situation in which monitoring is done;
- The information must be available on time; and
- Information should be in a tabular form/format.

The information provided for monitoring should not be seen as a tool to be used against the reporting officer for admonishing him. It is to be the basis for helping the officer through appropriate support and decisions. While monitoring certain aspects of a project all the related information must be available. For instance, in the case of monitoring the poultry complex project, the Block Development Officer should have information on the number of beneficiaries who entered the project, number of farms producing eggs, availability of marketing facilities, supply of feed for the birds and availability of veterinary services. If all the related information is provided, the situation can be reviewed in totality to help in taking appropriate decisions.

Timely availability of information is a prerequisite for effective monitoring. This may be every fortnight, every month or every quarter. The information should be available within a week after the reference period ends. Any delay in arrival of information may result in ineffective monitoring, thereby, leading to delays and reduction in benefits from the project.

One of the reasons why, in the existing monitoring system, the District Administration finds it difficult to use the information is the length of the

report. If it runs into a number of pages, no one would have enough time to read and use it. Therefore, it is suggested that the information provided should be in the form of simple tabular statements accompanied by brief comments from which the manager can know what has happened in the project, why certain results have not been achieved.

The analysis of the information should be simple. Percentage and ratio computed from the data provided in the report for monitoring can be effectively used for assessing the progress made in the project. The conclusions drawn should have a direct bearing on the analysis and should lead to a decision. Unless the decision is implemented, it may not be viewed as a logical culmination of the monitoring process. It is necessary to follow up the project after mid-course corrections are introduced so as to keep the project continuously under control during its execution phase.

12.3.2 Monitoring Time Schedule Adherence

Let us illustrate the method of monitoring the project to see whether the activities are going according to the given time schedule. For this, we use the schedule worked out for the occurrence of events as discussed. For each event, we have worked out the earliest occurrence time (EOT) and the latest start time (LST). These are what we have planned. While monitoring progress in the execution phase the actual dates on which the various events have occurred need to be made available. Against the planned schedule, the Block Development Officer can see whether the various events have occurred or not. The analysis of information in this context enables updating the schedule of implementation. After updating, the manager may come to the conclusion that the project is going according to the given schedule or is behind or ahead of the schedule. If the manager concludes that the events have occurred as per the schedule, there is no need for anxiety. But if the events have occurred behind the time schedule, apprehension would be that the project is going to be delayed. Wherever delays are seen, the decisions should be to compress the remaining portion of the project execution period. This is known as ‘crashing’ or compression.

- i) **Proforma for Reporting:** The information required for monitoring the occurrence of events can be given in the proforma below:

Proforma for Reporting on the Occurrence of Events

Event's Name	Earliest Occurrence Time (EOT)	Latest Occurrence Time (LOT)	Slack (3)-(2)	Actual Occurrence Time	If in Progress, Expected Occurrence Time	Reasons for delay, if any
(1)	(2)	(3)	(4)	(5)	(6)	(7)
1						
2						
3						

4						
5						
6						
7						

The event code numbers are provided in the first column. The second and third columns provide the planned occurrence time of events. The next column (column 4) provides the cushion available for occurrence of each event. This is the SLACK. In the first four columns, the planned schedule of events for occurrence is provided. The persons responsible for realizing each event in the project should report from column 5 onwards. In column 5 the actual date on which the event occurred has to be reported. At the time of reporting, certain activities may be in progress, as a result of which the events would not have occurred. In such cases, the persons responsible for the events will be required to indicate when the event is expected to occur.

The project activities may be delayed due to various reasons, some of which might be unavoidable. These would not have been anticipated at the time of formulating the project schedule. For example, the construction of poultry sheds could have been planned during the period of November to February. During this period, the labour may be in short supply, since they are engaged in agricultural operations. On account of this reason, the construction on poultry complex would get affected. Therefore, the Project Manager should know the reasons for any delay in occurrence of certain activities or events in the project. The information on the reasons for delay needs to be provided in column 7 of the proforma presented above. This information will help the Manager to take suitable steps to overcome delays.

The Project Manager should receive suggestions from the persons engaged in various activities, so that while taking decision these could be put to use. The suggestions are to be provided in the last column of the proforma given above.

- ii) **Time Schedule Updating:** The analysis of information is done by posting the data on actual occurrence time in the original schedule. If the actual occurrence time of non-critical events is less than the latest occurrence time, the project will not be delayed on account of this event. However, if the actual occurrence time exceeds the latest occurrence time of the event in the case of critical events, then the project will be delayed. By updating the schedule, the Manager may arrive at the conclusion on the likely completion of the overall project, whether it is going to be in time or behind time. The above method of monitoring progress and adherence to the time schedule is known as **network-based monitoring**.

- iii) **Monitoring Results or Benefit Flow:** Besides monitoring adherence to

a given time schedule, the Project Manager will have to see whether or not the expected results are generated according to the planned targets. This method of monitoring is known as information based monitoring system. The information required to monitor this aspect can also be provided in tabular form. The manager can, then, quickly review and take necessary managerial decisions for mid-course corrections. Let us take, for example, the project, which aims at alleviation of poverty in the villages through IRDP. The size of population below the poverty line is known from the survey conducted prior to launching the programme. The overall objective of IRDP is to help the families to come up the poverty line. For instance, in a village where there are say, 650 families, the programme would have covered, say, 250 families under IRDP assistance. The percentage of population covered to the total target is the coverage indicator, which gradually increases each year.

The Block Development Officer, while closely following the progress, is expected to have information on the number of families covered under the programme, which have crossed the poverty line. Out of the 225 families covered, in the earlier example, about 105 have crossed the poverty line. A ratio of the total number of families covered to those who have crossed the poverty line through the programme is an indicator of poverty alleviation. Though 105 families would have crossed the poverty line, all of them may not use the increased income for improving their socio-economic conditions. The information from the village on the number of such families, who have invested their surplus income to educate their children, improve their housing, provide better clothing and improve food habits, will be extremely useful in assessing the level of development brought about in the village.

12.3.3 Project Management Information System (MIS)

The information for assessing the performance of development projects can be obtained through the information system at the Block level. For designing an information system for monitoring, the following steps are to be followed:

- List the key data required to assess the performance of development projects. Take the example of poverty alleviation under IRDP. For this, the Block level organization would require information on the number of beneficiaries given loan, number of families whose income generating economic activities have been grounded, number of families earning more income than what they were getting prior to entering the programme, etc.
- Specify the channel through which the information should flow. In the case of Block level monitoring, the information should be collected at the grassroot level by the Village Level Worker (VLW) and reported to the Block Development Officer.

- Decide how often the information should be reported from one level to another, i.e., determine the periodicity. Some information may be required once in a year, while some others may be needed once in a quarter or at more frequent intervals.
- Develop appropriate formats in which information should be recorded and reported. The VLW will report in a format for his village, which should be consistent with other formats being used, for example, In a report by the extension officer to the Block Development Officer. It has to be designed in such a way that the person responsible for reporting may spend minimum time to furnish the data.
- Train the staff responsible for recording information and reporting the same to higher authorities. Unless training is imparted, the reliability of reporting, its regularity and punctuality cannot be ensured.

12.3.4 Monitoring System

A monitoring system forms a vital part of the management process. The degree of monitoring would depend upon the nature of activities. For example, critical activities would need close follow-up. The change in the critical path, which might occur frequently, also needs to be monitored very closely. For instance, equipment expected in two months' time might take six months due to a strike at the supplier's factory. The activity of installing the equipment might have been a non-critical activity. But with the fresh information, it could become a critical activity. Changes of this type underline the need for periodic reviews and updating of the project network. This forms the core of a monitoring system based on a sound Management Information System (MIS).

12.3.5 Review and Updating

Consider Figure 2 below. It represents the network for a project

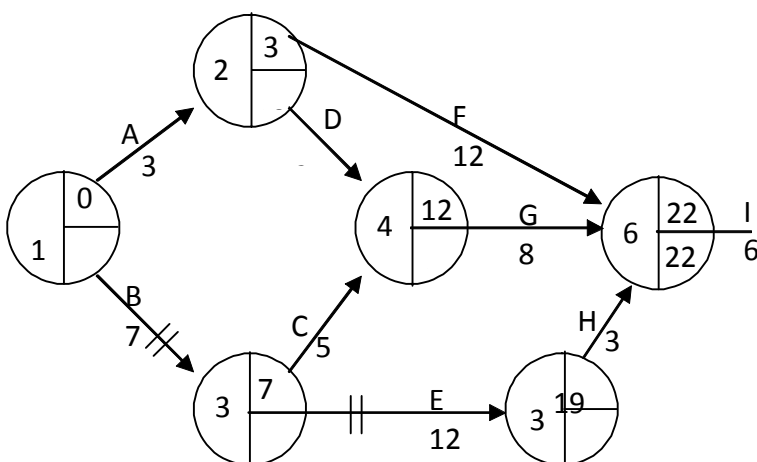


Fig. 2

A review was carried out at the end of the twelfth day. The progress, which should have been made and the progress reported is given below.

Progress as per original plan	Progress reported
Activities A and B completed or in progress and to take maximum two days to be completed.	A completed in 6 days, D completed by the end of 7 th day ahead of another schedule. B very much delayed, now expected to take another 8 days.
E and F in progress; G, H and I yet to be started	F is in progress and will take another 6 days. C, E, G, H and I are yet to be started

According to the first evidence, activities C and H are likely to take longer time than originally planned, i.e., 7 and 6 days respectively. There is no change in the estimated duration of other activities. It now requires updating of the course of action. How does one go about it? Well! A fresh network, updated one, needs to be prepared as per the procedure outlined below:

- Fix the earliest event time and latest event time of event (node) 1 at 12.
- For the completed activities, indicate the duration as zero.
- For the activities, which are in progress, indicate the durations equal to the time required for completion of the remaining portion of each activity.
- For the activities, which are yet to commence, indicate the durations according to the revised estimates.
- Complete the network calculations and find the critical path.

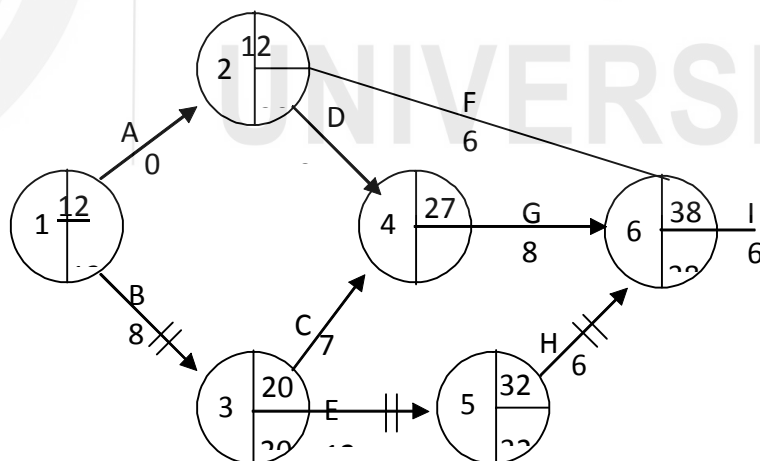


Fig. 3

Thus, the revised project completion time is 44 days and there is no change in the critical path. The periodicity of updating depends on the nature of the project. A large project generally requires less frequent updating as compared to a short duration project. Normally, with the progress of the project the frequency of updating increases to make all possible efforts for minimizing the effects of deviations of individual activities from the original schedule.

Check Your Progress 2

Note: a) Use the space given below.

b) Compare your answer with the text.

- 1) List two essential features of monitoring.

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- 2) What criteria should be met by a sound project monitoring information system?

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12.4 EVALUATION OF RURAL DEVELOPMENT PROJECTS

Since many developing nations seek to achieve growth with social justice through planning, the evaluation of development policies and programmes assumes importance for the following reasons:

- The social and economic structures are so complex that the planning process must take these into consideration and formulate strategies and programmes on the basis of certain assumptions. Evaluation enables the policy makers to know, to what extent the planning process was responsive to development needs. It has been noticed, for instance, that some of the development programmes failed owing to wrong identification of the problems, incorrect assumptions, improper formulation or inefficient administering of programmes.
- The resources at the command are limited. Evaluation will enable allocation of funds to programs. Selection of programmes will be guided by efficiency and cost-effective considerations.
- Evaluation aids decision makers to assess the various programmes in terms of the irrelevance (to overall development goals), efficiency, effectiveness and impact. This helps in better formulation and effective implementation of programmes in the future.

- The evaluation process is expected to make recommendations on the need for modification of an existing programme, charting of a different course of action, or formulation of new programmes. Thus, evaluation, in a way, helps to assess the long-term implications of the present decisions and programme choices.

More specifically, the objectives of evaluation are:

- Measuring the progress in both quantitative and qualitative aspects in relation to the stated objectives, targets and course of action;
- Assessing efficiency of the process of conversion of inputs in to outputs and the efficiency of the management apparatus;
- Identifying factors responsible for speeding up or slowing down of the process; that is, identifying reasons for a project's success or failure and
- Providing feedback to decision makers, which will enable modifications, where necessary, in the policy, strategy and objectives of the programme and the operational aspects.

12.4.1 Dimensions

Evaluation is now considered to be an integral part of the overall planning system and a main ingredient of development administration. Or, evaluation is a performance or achievement audit, which assesses the whole gamut of activities associated with programme formulation, implementation and impact, and compares the observed results with those anticipated. In other words, evaluation critically examines:

- Were the assumptions of the programme/project design valid?
- Were the programme objectives linked with specific schemes to achieve them?
- Were the objectives quantified in the form of targets?
- Did the implementation adhere to the prescribed design?
- Were the critical inputs delivered in time and in right quantity and quality by the agencies concerned?
- Was the manpower delivering the services trained?
- Was the scheduled path followed? If not, what were the reasons?
- Are the benefits commensurate with the costs?
- Has the intended impact occurred?
- Have the effects been beneficial or harmful?

It is obvious that answers to these questions indicate neither success nor failure of the project, but point out the problems that affect its implementation.

12.4.2 Major Concerns of Evaluation

Let us now discuss major concerns of evaluation.

1) Assessment of Outcome

Evaluation is concerned with assessment of outcome. This occurs after the completion of the programme. Some issues considered in outcome evaluation are:

- Whether the beneficiaries have realized the expected benefits.
- Whether beneficiaries are better off compared to non-beneficiaries on the relevant indicators.
- To what extent the programme/project is responsible for the observed benefits (both intended and unintended) and what have been the costs?
- What are the changes noticed in the behavioral dimensions of the target population?
- Have there been any adverse, unintended effects of the project?
- Have any forces been set in motion, the dynamics of which may have far reaching consequences?
- The discussion of causation is one of the major aspects of an evaluation. The isolation of the programme effects from those of other programmes is the real task involved in this exercise and demands a high level of expertise in the evaluation design.

2) Measurement of Project Efficiency

Sometimes, major concern of evaluation is the measurement of project efficiency. Two yardsticks used to measure project efficiency are:

- Cost efficiency expressed in terms of outcome per unit of cost; and
- Project effectiveness measured in terms of outcome per unit of effort.

For example, if two projects A and B are equally effective, but A is cost efficient, then the decision rule is to prefer A. But when A is more effective, but not cost efficient, then the criterion for projects election is whether the additional benefit is worth the additional cost. Another situation, which makes evaluation task rather difficult, is when projects have several characteristics. Each one brings improvements in certain characteristics. The assignment of weight on an objective basis to the progress made becomes the main focus. Arriving at a composite measure for judging relative effectiveness and efficiency of the individual is a major problem.

3) Purpose-Oriented Evaluation

Sometimes, evaluation is purpose-oriented. In this type of evaluation, the focus is on the basic purpose for which the projects have been launched. For example, the thrust of rural development programmes is to improve the

quality of life in rural areas. There may be several sub-purposes, such as provision of basic amenities to rural population. Two important types of evaluation in this category are:

- 1) Formative evaluation and
- 2) Summative evaluation

Formative evaluation is basically concerned with the formulation of projects. It is service-oriented and aids planners through identification of potential problem areas where project needs improvement. In other words, the purpose is to improve the planning design with a view to enhance the flow of benefits from the project or to increase the efficiency of the delivery system.

Summative evaluation aims at selection of projects from a set of projects for deciding the continuance or termination of the project. In this sense, there is finality to summative evaluation, since the evaluation report may recommend withdrawal of the project. In brief, it provides a summary statement about the effectiveness of the programme—incorporating the project's achievement of pronounced goals, unanticipated outcomes and comparison with alternative projects.

4) **Prerequisites for Evaluation**

For success of evaluation study, certain basic requirements have to be met. A few are given below.

- i) One of the foremost requirements is clearly specified objectives of the project and their likely impact. Their vagueness makes the task of evaluation difficult, as evaluation seeks to assess the extent of realization of the objectives. There may be a hierarchy of immediate, intermediate, final and ultimate objectives. A project description in terms of this hierarchy not only facilitates better formulation, but also helps in visualizing the project as an activity of interrelated sequences. The corresponding items in the project cycle are – inputs, outputs, effects and impact.

The immediate objective may be provision of inputs of right quality in adequate quantity in time to the programme clients. The 'inputs may as well include services. For example, in the agricultural extension programme, training of farmers by the extension worker is an input.

The intermediate objective could be generation of initial results (outputs) of certain order as a sequel to the utilization of inputs. Adoption of improved practices of agriculture by (contact) farmers is the immediate objective for the agricultural extension programme.

The final objective is the same as the project objective and is usually referred to as effects. The increase in yield (or incremental yield) is the effect measured in quantitative terms.

The ultimate objective relates to sustenance of the effect of the project and, thus, is concerned with the impact of the programme. For example, modernization of agriculture could be an ultimate objective of agricultural programmes.

- ii) Another prerequisite for evaluation is the establishment of norms/targets for comparison of results. In this context, it is useful to have bench-mark data or base line information relating to pre-launching (programme) period. Also, relevant are a set of indicators for measuring the extent of progress and, thus, determining the success or failure of the programme / project. Needless to say, the selected indicators should relate to the various objectives of the project and/or the various dimensions of these objective(s).
- iii) Equally important is the selection of techniques for data collection and analysis to make the findings of the study more meaningful.

12.4.3 Techniques and Criteria

You will find a wide range of evaluation designs—from simple to very complex and sophisticated ones. The application of various techniques depends on several factors such as time, cost, nature / type of questions the evaluation seeks to answer, and the expertise available. Use of scientific method is desirable since it involves careful observation and also controls the subjective bias of evaluators. The validity of inferences and findings of the study is greater if techniques are scientific. These methods ensure:

- Accurate and reliable database, as the tools used for collection of data and also the data collection procedures and sampling design are efficient, and
- The analytical techniques used help in disaggregating the whole complex phenomenon, and, thus, assessing the programme effects in a non-biased and objective manner.

12.4.4 Criteria for Evaluation

The following five criteria are usually adopted in evaluation studies:

- Efforts
- Performance
- Adequacy
- Efficiency
- Process

Efforts: Summarily, efforts refer to the total inputs to the programme. For example, a criterion relating to the amount of effort put into the programme would include quantity and quality of programme inputs (range of services), the coverage of population, programme personnel, financial resources, etc.

Performance: This deals with the consequences of the programme inputs. The composite index of performance should consider the magnitude and quality of benefits and changes in behavioural dimensions. Thus, it refers to output generated from the system as a result of programme inputs.

Adequacy: This is a relative measure indicating the relationship of the effort and performance of the programme to the level of need for the programme. That is, adequacy is the extent of successful (or effective) coverage of beneficiary population in the total population. It, thus, refers to the relationship between output and the total need.

Efficiency: This measure connects the above three criteria-effort, performance, adequacy and is based on the minimax principle, i.e., minimization of effort and maximization of performance. In other words, it is simple input-output ratio. Naturally, the various inputs, such as money, time and staff are being considered in evaluating the various alternatives.

Process: It is the study of the means of the programme in producing outputs (results) with a view to establishing causality.

12.4.5 Indicators of Impact

Indicators are simple measures reflecting a problem or changes in a situation. They are used as instruments for evaluating changes resulting from the various activities/efforts of the programme. Impact indicators are designed to show the degree of achievement of programme objectives. For example, the percentage of beneficiary households crossing the poverty line once for all is one of the (impact) indicators for the success of IRDP.

- i) **Characteristics of Indicators:** To be effective measures for assessing impact/effect indicators should possess the following characteristics:

Validity: What is measured is the same as what is intended to be measured. For instance, marks obtained by a student are a valid measure of scholastic achievement. It is, however, not a valid measure of his behaviour in school.

Reliability: Same values should be obtained even when the evaluation / data collection is done by others under similar conditions. In other words, repeated collection of data should yield the same values.

Relevance: Indicators selected should be related to programme objectives.

Sensitivity: Indicators should be able to capture even small changes taking place in important aspects.

Objectivity: Indicators should be free from subjective judgments of the evaluator.

Simplicity: Measurement should not be a costly or time-consuming affair.

The above discussion makes it clear that indicators selected should relate to the major dimensions of the programme. For instance, in case of service-oriented programmes, the coverage (physical), range of service and quality of service(s) rendered should be reflected by the selected indicators.

In case the goals are many and are heterogeneous in nature, separate indicators are to be considered. These indicators should be pivotal / key indicators and aid the programme authorities to take remedial measures wherever the programme path deviates from the planned trajectory. This also reduces the workload of officials in data collection.

- ii) **Differential Impact:** Indicators of impact, which are generally based on time series data, reflect direction, speed and extent of change. The interregional and intergroup comparisons help assess the differential impact of the programme. These indicators may be direct or indirect in nature. The former is amenable to quantification and are more easily measurable. The indirect indicators are those which are neither quantifiable nor cost-effective. In such cases, proxy indicators (partial measures) are usually employed. For example, an assessment of the nutritional status of children utilizes the 'age-weight' indicator as a proxy indicator. Similarly, partial measures are considered to capture behavioural changes.

Impact Indicators of Programmes: The comparison of indicator values – before and after programme of 'with and without' programme – will provide an estimate of the magnitude of achievement. The indicators considered in this context, for instance, are: impact of the programme on household income, use of incremental incomes (consumption, savings and investment) and the living standards.

A few impact indicators are suggested for some programs as an illustration.

Agricultural Programmes: Incremental yields, economic returns, employment generated.

Health and Nutrition Programmes: Morbidity rate, death rates, infant mortality rate, birth weight, weight for age, etc.

These indicators can be collected to different social groups to know the trends in the incidence and prevalence among these groups.

One of the most common evaluation methodologies for measuring impact is the 'before and after'. In this, base line data (relevant to the change being sought through the project) is collected before a project is launched. After the project has been completed (or some phase of it), the data are again collected. The difference is, then, interpreted in terms of project objectives, operations and impact.

Another commonly used evaluation methodology is the experimental group – control group design (i.e., with and without programme). Two population groups are selected and matched. In one, the project is implemented (the experimental group). In the other, no such project is under implementation (the control group). The two are, then, compared on relevant indicators. The difference is interpreted in terms of project objectives, operations and impact.

iii) **Miscellaneous:** For any programme, the following four interrelated indices of impact are suggested:

- 1) **Goal Effectiveness Index (GEI):** Indicates the extent of achievement of goals/ objectives, i.e., comparison of programme's result with the objectives. It answers the question whether the programme accomplished its objectives or not.

$$\text{GEI} = \text{Programme Result} / \text{Programme Task Objective}$$

- 2) **Potency Effectiveness Index (PEI):** It points out the achievement as a proportion of the total need. For instance, in the case of IRDP, it refers to the number of beneficiaries who have crossed the poverty line to the number of poor who need such programme assistance.

$$\text{PEI} = \text{Programme Result} / \text{Estimated size of the potential client needed for the project}$$

- 3) **Realized Efficiency Index (REI):** It compares the result with the total effort directed towards the programme and it is similar to the output-input ratio.

$$\text{REI} = \text{Programme Result} / \text{Total Programme Efforts}$$

- 4) **Capacity Efficiency Index (CEI):** Compares programme outcome with potential capacity of the programme in rendering its services as defined by its physical, human and financial resources. How much the programme can accomplish with the resources at its command is the issue under consideration.

$$\text{CEI} = \text{Programme Result} / \text{Established Potential Capacity of Programme}$$

12.4.6 Concurrent and Terminal Evaluation

You may have come across the terms 'concurrent evaluation' and 'terminal evaluation'. Concurrent evaluation is an on-going activity and is carried out when the programme is under implementation. It is usually undertaken at regular intervals or at the end of a phase, to give an objective feedback on the progress and direction of the programme. Concurrent evaluation does not provide a feedback on impact, unless it is at the end of a phase when some impact was visualized in the project design. The utility of concurrent evaluation is that it can assist decision makers in modifying the objectives (if necessary), and change the implementation strategy, so that overall objectives

of the project can be achieved. Concurrent evaluation also permits examination of the relevance or validity of the assumptions sufficiently before the terminal stage.

Thus, the main purposes of concurrent evaluation are:

- To have a feedback on the need for modifying the project in relation to the changes in the project environment or in the development goals, and
- To identify the problems affecting the program's success with a view to provide corrective action.

Concurrent evaluation is an internal function of project managers who are also the users of the information provided by the evaluation. Concurrent evaluation can, thus, be viewed as a built-in 'learning dimension' of the programme/ project.

Terminal evaluation is done at the end of programme implementation and focuses on the impact and other complex issues of programme performance.

Check Your Progress 3

Note: a) Write your answers in the space provided.

b) Compare your answers with the text.

- 1) Why evaluation is considered an essential tool of development administration?

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- 2) What are major concerns of evaluation?

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- 3) Is concurrent evaluation more important than terminal evaluation?

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12.5 LET US SUM UP

Management is the art of getting things done by directing the efforts of other people so as to bring about the best possible allocation and utilization of human, material and financial resources for attainment of goals. Since rural development setting consists of people, area and activities their understanding is a prerequisite for effective management. One of the main functions of management is effective execution/ implementation of programmes. Implementation is a process by which a set of predetermined activities are carried out in a planned manner to achieve certain established objectives. Monitoring is an important management tool for overseeing the implementation of a project through information feedback on the direction and pace of the project, so as to take timely corrective action, if required. Evaluation is another integral part of the overall planning and implementation system. Evaluation provides a feedback to the policy makers and enables them to judge the effectiveness of strategies, thus, helping them to modify or work out alternative strategies and programmes. Evaluation enables not only measurement of progress in qualitative and quantitative terms, but also assesses the impact and efficiency of the implementation machinery and the identification of factors, which obstruct development.

12.6 KEY WORDS

Implementation: Process of carrying out a set of predetermined activities in a planned manner to achieve certain stated objectives.

Information System: An integrated system consisting of collection, compilation, analysis and use of data pertaining to the project.

Monitoring: Management device to detect deviation from the planned action path and to take corrective action.

Evaluation: Performance or achievement audit that assesses the programme formulation, implementation and impact, and compares the observed results with the anticipated ones.

12.7 REFERENCES AND FURTHER READINGS

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